

Aejandro De La Torre

LIS 440

Lecture: *wk11: Lecture-Demo: State Housing Data*

11/11/25

Code

```
# Import the necessary packages
import pandas as pd # because 6 characters is to much
import os
from pathlib import Path # to set up paths easier

# Print version of pandas
print(pd.__version__)

# Set working directory (CWD) : root folder
root = Path(__file__).parent.parent.resolve()

# Show current working directory
print("Current Working Directory: ",Path.cwd())

# Set CWD: Data
data = root / 'data'
# Set CWD: Data > Raw
data_raw = data / 'raw'
# Set CWD: Data > Processed
data_processed = root / 'data' / 'processed'
# Set CWD: Scripts
scripts = root / 'scripts'

# Read Student Data
wisc_data = data_processed / 'StateHousing2023_wisconsin.csv'

# Create dataframe of student data
wisc_df = pd.read_csv(wisc_data) # He labeled it wisconsinHousingDF but too long

# View first and last 3 entries
print("First 5 rows: \n", wisc_df.head(3)) # just show head
print("Last 5 rows: \n", wisc_df.tail(3)) # check if I am there

# Ask for shape of Function
print("Shape of data: \n", wisc_df.shape)
print("Columns:", wisc_df.columns )

# Get the number of rows and columns using .shape
num_rows, num_cols = wisc_df.shape
print(f"Total number of rows: {num_rows}")
print(f"Total number of columns: {num_cols}")
```

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# Trying out summary stat funcs from
https://pandas.pydata.org/docs/getting_started/intro_tutorials/06_calculate_statistics.
html
print(wisc_df.describe())

# Following what prof does:
print("Last 2 rows: \n", wisc_df.tail(2))

valueColumnSeries = wisc_df["Value"]
print("Value Column: ", valueColumnSeries)

# Convert the Value column from String to Numeric
# 1. Remove the '$' and ',' characters
wisc_df['Value'] = wisc_df['Value'].str.replace(r'[$,]', '', regex=True)
# 2. Convert the cleaned column to a numeric type
wisc_df['Value'] = pd.to_numeric(wisc_df['Value']) # numeric to then be able to
calculate mean, etc
print(wisc_df.head(1)) # check if convert successfull

# Print the corrected col value entries
print("The Values column has ", valueColumnSeries.size, "entries.")

# Basic summary stats
print("Average Value (Mean):", wisc_df['Value'].mean())
print("Median Value:", wisc_df['Value'].median())
print("Mode Value(s):\n", wisc_df['Value'].mode())

# EDITS PER FEEDBACK
# Missing: (1) print your name, (2) print the whold df
print("Alejandro De La Torre's code...") # fixes prob 1
print(wisc_df) # fixes prob 2

print("End of program.")

```

Shell

```

>>> %Run 11-11_thonny_demo.py
2.3.3
Current Working Directory: /Users/alexdelatorre/Library/CloudStorage/OneDrive-UW-Madison/Coursework/L I S
440/Lectures/Week 11/Tuesday/scripts
First 5 rows:
   Month    Value
0 1/31/23 $268,736.68
1 2/28/23 $270,601.00
2 3/31/23 $273,370.26
Last 5 rows:
   Month    Value
9 10/31/23 $285,186.25
10 11/30/23 $285,404.52

```

11 12/31/23 \$285,129.45

Shape of data:

(12, 2)

Columns: Index(['Month', 'Value'], dtype='object')

Total number of rows: 12

Total number of columns: 2

	Month	Value
count	12	12
unique	12	12
top	1/31/23	\$268,736.68
freq	1	1

Last 2 rows:

	Month	Value
10	11/30/23	\$285,404.52
11	12/31/23	\$285,129.45

Value Column: 0 \$268,736.68

1	\$270,601.00
2	\$273,370.26
3	\$276,247.66
4	\$279,194.59
5	\$281,146.74
6	\$282,099.93
7	\$283,201.35
8	\$284,443.19
9	\$285,186.25
10	\$285,404.52
11	\$285,129.45

Name: Value, dtype: object

	Month	Value
0	1/31/23	268736.68

The Values column has 12 entries.

Average Value (Mean): 279563.4683333333

Median Value: 281623.33499999996

Mode Value(s):

0	268736.68
1	270601.00
2	273370.26
3	276247.66
4	279194.59
5	281146.74
6	282099.93
7	283201.35
8	284443.19
9	285129.45
10	285186.25
11	285404.52

Name: Value, dtype: float64

Alejandro De La Torre's code...

	Month	Value
0	1/31/23	268736.68
1	2/28/23	270601.00
2	3/31/23	273370.26
3	4/30/23	276247.66

```
4 5/31/23 279194.59
5 6/30/23 281146.74
6 7/31/23 282099.93
7 8/31/23 283201.35
8 9/30/23 284443.19
9 10/31/23 285186.25
10 11/30/23 285404.52
11 12/31/23 285129.45
```

End of program.

```
>>>
```

